

# **EHR WORLD WIDE**

## **The Meaningful Use (MU) Program in the USA**

- The “Meaningful Use” Regulation for Electronic Health Records
- Does the meaningful use of electronic health records improve patient outcomes?
- Unintended Consequences of Nationwide Electronic Health Record Adoption: Challenges and Opportunities in the Post-Meaningful Use Era

- <2009: Use of EHRs is very limited in the USA
- Largest EHR system is the VistA system of the **Veterans Affairs Administration**
  - Services (including health care) for war veterans and their families
  - [https://en.wikipedia.org/wiki/United\\_States\\_Department\\_of\\_Veterans\\_Affairs](https://en.wikipedia.org/wiki/United_States_Department_of_Veterans_Affairs)
  - [https://de.wikipedia.org/wiki/Kriegsveteranenministerium\\_der\\_Vereinigten\\_Staaten](https://de.wikipedia.org/wiki/Kriegsveteranenministerium_der_Vereinigten_Staaten)

**Table 1. Veteran Population, VA Enrollees, and VA Patients, FY2001–FY2014**

| Year   | Total Veteran Population | VA-Enrolled Veterans | Patients Using VA Health Care During the Year |              |                |
|--------|--------------------------|----------------------|-----------------------------------------------|--------------|----------------|
|        |                          |                      | Veterans                                      | Non-Veterans | Total Patients |
| FY2001 | 26,092,046               | 5,124,168            | 3,890,871                                     | 356,333      | 4,247,204      |
| FY2002 | 25,627,596               | 6,248,949            | 4,290,717                                     | 380,320      | 4,671,037      |
| FY2003 | 25,217,342               | 7,186,643            | 4,544,430                                     | 417,023      | 4,961,453      |
| FY2004 | 24,862,857               | 7,419,851            | 4,713,583                                     | 453,250      | 5,166,833      |
| FY2005 | 24,521,247               | 7,655,562            | 4,862,992                                     | 445,322      | 5,308,314      |
| FY2006 | 24,179,183               | 7,872,438            | 5,030,582                                     | 435,488      | 5,466,070      |
| FY2007 | 23,816,018               | 7,833,445            | 5,015,689                                     | 463,240      | 5,478,929      |
| FY2008 | 23,442,489               | 7,834,763            | 5,078,269                                     | 498,420      | 5,576,689      |
| FY2009 | 23,066,965               | 8,048,560            | 5,221,583                                     | 523,110      | 5,744,693      |
| FY2010 | 23,031,892               | 8,343,117            | 5,441,059                                     | 559,051      | 6,000,110      |
| FY2011 | 22,676,149               | 8,574,198            | 5,582,171                                     | 584,020      | 6,166,191      |
| FY2012 | 22,328,279               | 8,762,548            | 5,680,374                                     | 652,717      | 6,333,091      |
| FY2013 | 21,972,964               | 8,926,546            | 5,803,890                                     | 680,774      | 6,484,664      |
| FY2014 | 21,619,731               | 9,111,955            | 5,908,042                                     | 708,921      | 6,616,963      |

- <https://www.fas.org/sgp/crs/misc/R43579.pdf>

- 2016: 8.7 Million patients
- 1700 sites
  - Medical centers
  - Outpatient clinics
  - Community service programs
  - Community based outpatient clinics

<http://www.va.gov/directory/guide/division.asp?dnum=1>

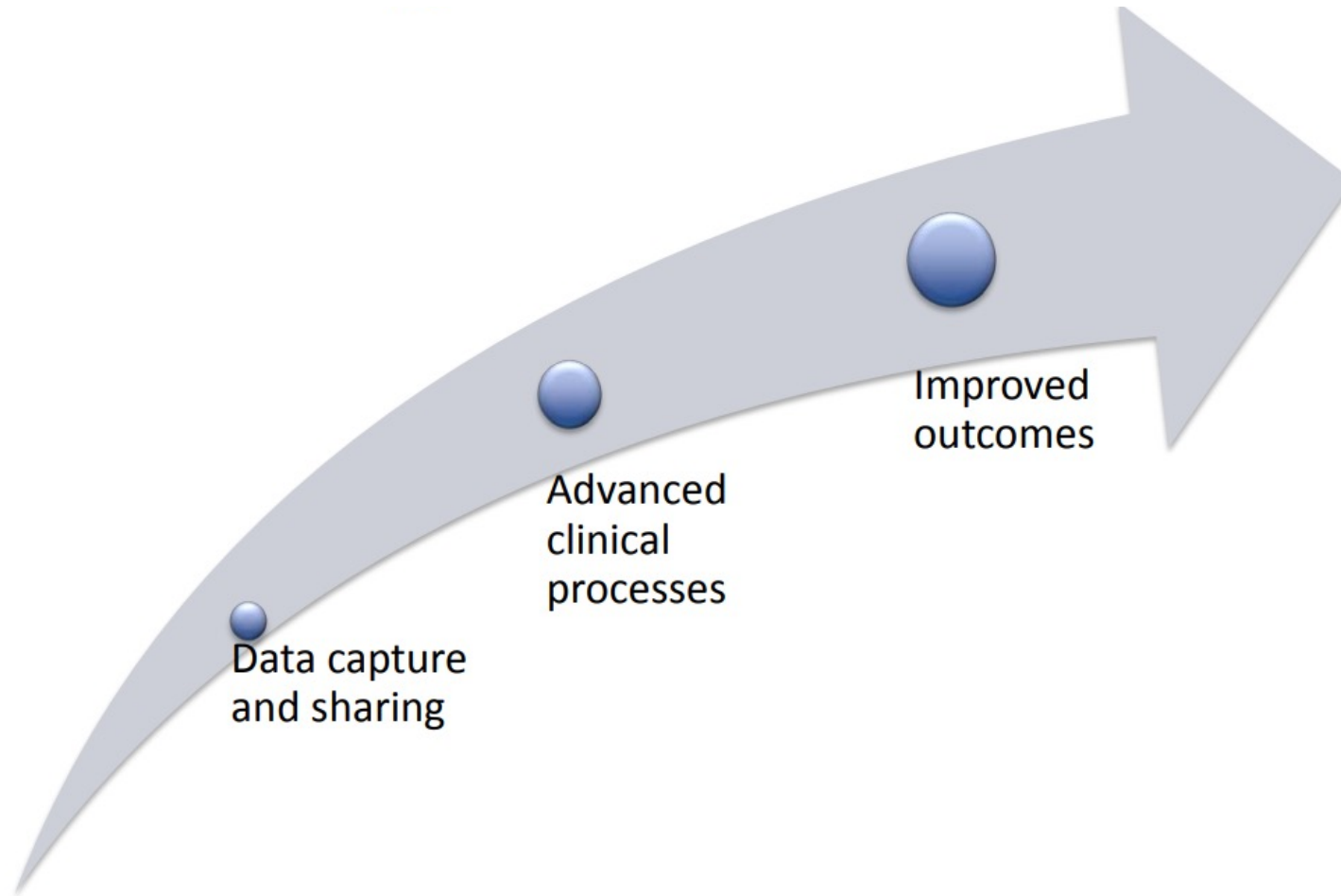
- Late 1970s: started as an underground movement
- Programmers at VA hospitals secretly developed a system to exchange medical information between different VA hospitals (the "[hardhats](#)")
- Original name: Decentralized Hospital Computer Program (DHCP)
- "Legalization" in 1981-1982
- Later renamed into VistA
- Available as "public domain" - also used in other countries

- American Reinvestment and Recovery Act (ARRA)
  - primary objective of this federal statute was to save existing jobs and create new ones as soon as possible
  - invest in infrastructure, education, health, and renewable energy
- The Health Information Technology for Economic and Clinical Health Act (HITECH Act)
  - motivate the implementation of electronic health records (EHR) and supporting technology in the United States
- **Incentives** for the use of Health Information Technology - budget 36.106 US\$
- 2009 was the year President Obama came in office

- Meaningful Use is using certified EHR technology to
  - Improve quality, safety, efficiency, and reduce health disparities
  - Engage patients and families in their health care
  - Improve care coordination
  - Improve population and public health
  - All the while maintaining privacy and security
- Meaningful Use mandated in law to receive incentives



- The Recovery Act specifies the following 3 components of Meaningful Use:
  1. Use of certified EHR in a meaningful manner (e.g., e-prescribing)
  2. Use of certified EHR technology for electronic exchange of health information to improve quality of health care
  3. Use of certified EHR technology to submit clinical quality measures (CQM) and other such measures selected by the Secretary



- Central government has limited power in the USA
- Unlike in Austria, imposing a central system would not work
- Therefore, a system of **incentives** was put into place
- Medicare and Medicaid providers receive money when they install EHR systems that meet some minimal requirements
  - **Medicare**: insurance programs for people >65 and younger disabled people.
  - **Medicaid**: assistance program for lower-income people
- <https://www.hhs.gov/answers/medicare-and-medicaid/what-is-the-difference-between-medicare-medicaid/index.html>

- Medicare providers can receive up to 44000 US\$
  - Medicaid providers can receive up to 63000 US\$
- 
- Use of certified EHR technology - in a meaningful matter
  - Use of certified EHR technology for electronic exchange of health information to improve quality of health care
  - Use of certified EHR technology to submit clinical quality measures (CQM)

- After 3 years, Medicare and Medicaid providers that do not meet the requirements get "payment adjustments" (i.e. get less reimbursement)
  - 1% first year
  - 2% second year
  - 3% third year

- 80% of the patients must have records in the certified EHR technology
  - Eligible Professionals must complete:
    - 15 core objectives
    - 5 objectives out of 10 from menu set
    - 6 total Clinical Quality Measures (3 core or alternate core, and 3 out of 38 from additional set)
  - Hospitals must complete:
    - 14 core objectives
    - 5 objectives out of 10 from menu set
    - 15 Clinical Quality Measures
- [https://www.cms.gov/Regulations-and-Guidance/Legislation/EHRIncentivePrograms/downloads/MU\\_Stage1\\_ReqOverview.pdf](https://www.cms.gov/Regulations-and-Guidance/Legislation/EHRIncentivePrograms/downloads/MU_Stage1_ReqOverview.pdf)



## Meaningful Use: Core Objectives

### • Eligible Professionals – 15 Core Objectives

Medication  
Prescription

1. Computerized provider order entry (CPOE)
2. E-Prescribing (eRx)
3. Report ambulatory clinical quality measures to CMS/States
4. Implement one clinical decision support rule
5. Provide patients with an electronic copy of their health information, upon request
6. Provide clinical summaries for patients for each office visit
7. Drug-drug and drug-allergy interaction checks
8. Record demographics
9. Maintain an up-to-date problem list of current and active diagnoses
10. Maintain active medication list
11. Maintain active medication allergy list
12. Record and chart changes in vital signs
13. Record smoking status for patients 13 years or older
14. Capability to exchange key clinical information among providers of care and patient-authorized entities electronically
15. Protect electronic health information

CMS = Centers for  
Medicare and Medicaid  
Services





## Meaningful Use: Core Objectives

- **Hospitals– 14 Core Objectives**

1. Computerized provider order entry (CPOE)
2. Drug-drug and drug-allergy interaction checks
3. Record demographics
4. Implement one clinical decision support rule
5. Maintain up-to-date problem list of current and active diagnoses
6. Maintain active medication list
7. Maintain active medication allergy list
8. Record and chart changes in vital signs
9. Record smoking status for patients 13 years or older
10. Report hospital clinical quality measures to CMS or States
11. Provide patients with an electronic copy of their health information, upon request
12. Provide patients with an electronic copy of their discharge instructions at time of discharge, upon request
13. Capability to exchange key clinical information among providers of care and patient-authorized entities electronically
14. Protect electronic health information



- For all of them see slide set at:

[https://www.cms.gov/Regulations-andGuidance/Legislation/EHRIncentivePrograms/downloads/MU\\_Stage1\\_Req\\_Overview.pdf](https://www.cms.gov/Regulations-andGuidance/Legislation/EHRIncentivePrograms/downloads/MU_Stage1_Req_Overview.pdf)

| Objective                                                                     | Measure                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EP Only: Generate and transmit permissible prescriptions electronically (eRx) | More than 40% of all permissible prescriptions written by the EP are transmitted electronically using certified EHR technology                                                                           |
| Maintain up to date problem list of current and active diagnoses              | More than 80% of all unique patients seen by the EP or admitted to the eligible hospital have at least one entry or an indication that no problems are known for the patient recorded as structured data |

EP = Eligible Provider

- 5 out of 10 must be reached

Reconciliation =  
Abstimmung

- **Eligible Professionals – 10 Menu Objectives**

1. Drug-formulary checks
2. Incorporate clinical lab test results as structured data
3. Generate lists of patients by specific conditions
4. Send reminders to patients per patient preference for preventive/follow up care
5. Provide patients with timely electronic access to their health information
6. Use certified EHR technology to identify patient-specific education resources and provide to patient, if appropriate
7. Medication reconciliation
8. Summary of care record for each transition of care/referrals
9. Capability to submit electronic data to immunization registries/systems\*
10. Capability to provide electronic syndromic surveillance data to public health agencies\*

\* At least 1 public health objective must be selected.

- 5 out of 10 must be reached

- **Hospitals– 10 Menu Objectives**

1. Drug-formulary checks
2. Record advanced directives for patients 65 years or older
3. Incorporate clinical lab test results as structured data
4. Generate lists of patients by specific conditions
5. Use certified EHR technology to identify patient-specific education resources and provide to patient, if appropriate
6. Medication reconciliation
7. Summary of care record for each transition of care/referrals
8. Capability to submit electronic data to immunization registries/systems\*
9. Capability to provide electronic submission of reportable lab results to public health agencies\*
10. Capability to provide electronic syndromic surveillance data to public health agencies\*

\* At least 1 public health objective must be selected.

- 2011:
  - 118,819 EPs received an average of US\$ 18,600
  - 2,320 eligible hospitals received an average of US\$ 1.6 Mio
  - Productivity losses estimated to be 20% for the first month of implementation
  - first year EHR-related revenue losses of US \$11,200 per EP
  - additional costs "associated with getting and learning a new system" estimated at US\$ 10,325 per EP



- 2012:
  - 268,461 EPs received an average of US\$ 15,200
  - 3,307 eligible hospitals received an average of US\$ 1.6 Mio
  - Total incentive costs 2011-2012 were US\$ 15.2 billion —roughly 42 percent of the initial US\$ 36 billion appropriated for Meaningful Use incentives
  
- 2013 (final year MU-1):
  - participation slowed and payments began to drop
  - 320,151 EPs and 4,215 hospitals

- More emphasis on the use of EHRs and **exchange**

## Electronic Health Records (EHR) Incentive Programs

The Stage 1 final rule set the foundation for the Medicare and Medicaid EHR Incentive Programs by establishing requirements for the electronic capture of clinical data, including providing patients with electronic copies of health information.

The Stage 2 final rule expanded upon the Stage 1 criteria with a focus on ensuring that the meaningful use of EHRs supported the aims and priorities of the National Quality Strategy. Stage 2 criteria encouraged the use of health IT for continuous quality improvement at the point of care and the **exchange of information in the most structured format possible.**

## EHR Incentive Payment Timeline

### Go Paperless and Get Paid

The American Recovery and Reinvestment Act of 2009 (Recovery Act) authorizes the Centers for Medicare & Medicaid Services (CMS) to award incentive payments to eligible professionals who demonstrate Meaningful Use of a certified electronic health record (EHR).

*It pays to get started early!* Providers must participate early — in 2011 or 2012 — in order to receive the maximum incentive.

These incentive programs will not always be available, and financial penalties are scheduled to take effect in 2015 for Medicare and Medicaid providers who do not transition to EHRs.

# Meaningful Use Stage 2: 2014-2016 - Medicaid

Medicaid EHR Incentive Payment Schedule for Eligible Professionals

|                          | Medicaid EP Qualifies to Receive First Payment in 2011 | Medicaid EP Qualifies to Receive First Payment in 2012 | Medicaid EP Qualifies to Receive First Payment in 2013 | Medicaid EP Qualifies to Receive First Payment in 2014 | Medicaid EP Qualifies to Receive First Payment in 2015 | Medicaid EP Qualifies to Receive First Payment in 2016 |
|--------------------------|--------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------|
| Payment Amount in 2011   | \$21,250.00                                            | \$0.00                                                 | \$0.00                                                 | \$0.00                                                 | \$0.00                                                 | \$0.00                                                 |
| Payment Amount in 2012   | \$8,500.00                                             | \$21,250.00                                            | \$0.00                                                 | \$0.00                                                 | \$0.00                                                 | \$0.00                                                 |
| Payment Amount in 2013   | \$8,500.00                                             | \$8,500.00                                             | \$21,250.00                                            | \$0.00                                                 | \$0.00                                                 | \$0.00                                                 |
| Payment Amount in 2014   | \$8,500.00                                             | \$8,500.00                                             | \$8,500.00                                             | \$21,250.00                                            | \$0.00                                                 | \$0.00                                                 |
| Payment Amount in 2015   | \$8,500.00                                             | \$8,500.00                                             | \$8,500.00                                             | \$8,500.00                                             | \$21,250.00                                            | \$0.00                                                 |
| Payment Amount in 2016   | \$8,500.00                                             | \$8,500.00                                             | \$8,500.00                                             | \$8,500.00                                             | \$8,500.00                                             | \$21,250.00                                            |
| Payment Amount in 2017   | \$0.00                                                 | \$8,500.00                                             | \$8,500.00                                             | \$8,500.00                                             | \$8,500.00                                             | \$8,500.00                                             |
| Payment Amount in 2018   | \$0.00                                                 | \$0.00                                                 | \$8,500.00                                             | \$8,500.00                                             | \$8,500.00                                             | \$8,500.00                                             |
| Payment Amount in 2019   | \$0.00                                                 | \$0.00                                                 | \$0.00                                                 | \$8,500.00                                             | \$8,500.00                                             | \$8,500.00                                             |
| Payment Amount in 2020   | \$0.00                                                 | \$0.00                                                 | \$0.00                                                 | \$0.00                                                 | \$8,500.00                                             | \$8,500.00                                             |
| Payment Amount in 2021   | \$0.00                                                 | \$0.00                                                 | \$0.00                                                 | \$0.00                                                 | \$0.00                                                 | \$8,500.00                                             |
| TOTAL Incentive Payments | \$63,750.00                                            | \$63,750.00                                            | \$63,750.00                                            | \$63,750.00                                            | \$63,750.00                                            | \$63,750.00                                            |



## Eligible Professional Core Objectives

Computerized Physician Order Entry (CPOE) for Medication, Laboratory and Radiology Orders

D1 - Improve Quality, Safety, Efficiency

Patient Ability to Electronically View, Download & Transmit (VDT) Health Information

D2 - Engage Patients & Family

Patient-Specific Education Resources

D2 - Engage Patients & Families

e-Prescribing (eRx)

D1 - Improve Quality, Safety, Efficiency

Clinical Summaries

D2 - Engage Patients & Family

Medication Reconciliation

D3 - Improve Care Coordination

Record Demographics

D1 - Improve Quality, Safety, Efficiency

Protect Electronic Health Information

D5 - Ensure Privacy & Security for Personal Health Information

Summary of Care

D3 - Improve Care Coordination

Record Vital Signs

D1 - Improve Quality, Safety, Efficiency

Clinical Lab - Test Results

D1 - Improve Quality, Safety, Efficiency

Immunization Registries

D4 - Improve Public & Population Health

Record Smoking Status

D1 - Improve Quality, Safety, Efficiency

Patient Lists

D1 - Improve Quality, Safety, Efficiency

Use Secure Electronic Messaging

D2 - Engage Patients & Families

Clinical Decision Support Rule

D1 - Improve Quality, Safety, Efficiency

Preventative Care

D1 - Improve Quality, Safety, Efficiency

**Core Objectives =  
Mandatory**

## Eligible Professional Menu Objectives

Syndromic Surveillance Data  
Submission

D4 - Improve Public & Population Health

Imaging Results

D1 - Improve Quality, Safety, Efficiency

Report Cancer Cases

D4 - Improve Public & Population Health

Electronic Notes

D1 - Improve Quality, Safety, Efficiency

Family Health History

D1 - Improve Quality, Safety, Efficiency

Report Specific Cases

D4 - Improve Public & Population Health

Menu objectives =  
Choices can be made to which ones  
to apply

## ■ Vital Signs

### Changes from Meaningful Use Stage 1:

|           | Stage 1                                                                                                                                                                                                                                                                | Stage 2                                                                                                                                                                                                                                                                                               |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Objective | <p>Record and chart changes in vital signs:</p> <ul style="list-style-type: none"><li>• Height</li><li>• Weight</li><li>• Blood pressure</li><li>• Calculate and display BMI</li><li>• Plot and display growth charts for children 2-20 years, including BMI</li></ul> | <p>Record and chart changes in vital signs:</p> <ul style="list-style-type: none"><li>• Height</li><li>• Weight</li><li>• Blood pressure <u>(age 3 and over)</u></li><li>• Calculate and display BMI</li><li>• Plot and display growth charts for patients <u>0-20 years</u>, including BMI</li></ul> |
| Measure   | <p>For more than 50% of all unique patients age 2 and over seen by the EP, blood pressure, height and weight are recorded as structured data</p>                                                                                                                       | <p>More than <u>80%</u> of all unique patients seen by the EP have blood pressure <u>(for patients age 3 and over only)</u> and height and weight <u>(for all ages)</u> recorded as structured data</p>                                                                                               |

## ■ Laboratory Results

### Changes from Meaningful Use Stage 1:

The Clinical Lab-Test Results Objective changed from being a Menu Objective in Stage 1 to a **Core** Objective in Stage 2.

|           | Stage 1                                                                                                                                                                                                                               | Stage 2                                                                                                                                                                                                                                             |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Objective | Incorporate clinical lab-test results into certified EHR technology as structured data                                                                                                                                                | Incorporate clinical lab-test results into Certified EHR Technology as structured data                                                                                                                                                              |
| Measure   | More than 40% of all clinical lab tests results ordered by the EP during the EHR reporting period whose results are either in a positive/negative or numerical format are incorporated in Certified EHR Technology as structured data | More than <b><u>55%</u></b> of all clinical lab tests results ordered by the EP during the EHR reporting period whose results are either in a positive/negative or numerical format are incorporated in Certified EHR Technology as structured data |



- Progress notes

## Electronic Notes

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### Objective:

Record electronic notes in patient records.

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### Measure:

Enter at least one electronic progress note created, edited and signed by an EP for more than 30 percent of unique patients with at least one office visit during the EHR reporting period. The text of the electronic note must be text searchable and may contain drawings and other content.

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### Clinical Importance:

Providers are encouraged to electronically record progress notes to provide access to the most well rounded patient information available to support continuity of care across patient care settings. Narrative entries in the progress note are an important component of patient records and complement data captured in defined structured fields; together these components create a more complete picture of the patient's status and can be used to track patient progress and sharing of information across care settings.

**Table 1: Initial Amount Calculation**

| Type of Hospital         | Hospitals with 1,149 or fewer discharges during the payment year | Hospitals with at least 1,150 but no more than 23,000 discharges during the payment year                                                     | Hospitals with 23,000 or more discharges during the payment year |
|--------------------------|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Base Amount              | \$2,000,000                                                      | \$2,000,000                                                                                                                                  | \$2,000,000                                                      |
| Discharge-Related Amount | \$0                                                              | $\$200 \times (n - 1,149)$ (n is the number of discharges (see above discussion of the discharge related amount for additional information)) | $\$200 \times (23,000 - 1,149)$                                  |
| Total Initial Amount     | \$2,000,000                                                      | Between \$2,000,000 and \$6,370,200 depending on the number of discharges                                                                    | Limited by law to \$6,370,200                                    |

## EHR Incentive Program Objectives and Measures for 2016 Table of Contents

Date updated: February 4, 2016

| Eligible Professional Objectives and Measures |                                                                                                                                                                                                                                                                                     |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (1)                                           | <a href="#">Protect electronic protected health information</a> created or maintained by the CEHRT through the implementation of appropriate technical capabilities.                                                                                                                |
| (2)                                           | Use <a href="#">clinical decision support</a> to improve performance on high priority health conditions.                                                                                                                                                                            |
| (3)                                           | Use <a href="#">computerized provider order entry</a> (CPOE) for medication, laboratory, and radiology orders directly entered by any licensed healthcare professional that can enter orders into the medical record per state, local, and professional guidelines.                 |
| (4)                                           | Generate and transmit permissible discharge <a href="#">prescriptions electronically (eRx)</a> .                                                                                                                                                                                    |
| (5)                                           | <a href="#">Health Information Exchange</a> - The eligible hospital or CAH who transitions their patient to another setting of care or provider of care or refers their patient to another provider of care provides a summary care record for each transition of care or referral. |
| (6)                                           | Use clinically relevant information from CEHRT to identify <a href="#">patient-specific education resources</a> and provide those resources to the patient.                                                                                                                         |
| (7)                                           | The eligible hospital, or CAH who receives a patient from another setting of care or provider of care or believes an encounter is relevant performs <a href="#">medication reconciliation</a> .                                                                                     |
| (8)                                           | <a href="#">Patient electronic access</a> - Provide patients the ability to view online, download, and transmit their health information within 36 hours of hospital discharge.                                                                                                     |
| (9)                                           | <a href="#">Public Health Reporting</a> - The eligible hospital or CAH is in active engagement with a public health agency to submit electronic public health data from CEHRT except where prohibited and in accordance with applicable law and practice.                           |

- [https://www.cms.gov/Regulations-and-Guidance/Legislation/EHRIncentivePrograms/Downloads/2016\\_EHTableOfContents.pdf](https://www.cms.gov/Regulations-and-Guidance/Legislation/EHRIncentivePrograms/Downloads/2016_EHTableOfContents.pdf)



- Optional for providers in 2017
- Mandatory for everyone in 2018
  
- Physicians:
  - 8 overall (core) requirements
  - No menu options anymore
  
- Hospitals
  - Depend on whether Stage 1 or Stage 2 has been reached

[https://www.cms.gov/Regulations-andGuidance/Legislation/EHRIncentivePrograms/Stage3Medicaid\\_Require.html](https://www.cms.gov/Regulations-andGuidance/Legislation/EHRIncentivePrograms/Stage3Medicaid_Require.html)

- More emphasis on interoperability and health information exchange
- MACRA
  - Medicare Access and CHIP Reauthorization Act of 2015 (CHIP = State Children's Health Insurance Program)
- Moving gradually from "pay for service" to "pay for performance"
  - providers are rewarded for improved health outcomes rather than specific services

## Value-Based Health Care Benefits

### PATIENTS

Lower Costs  
& better  
outcomes

### PROVIDERS

Higher Patient  
Satisfaction  
Rates &  
Better Care  
Efficiencies

### PAYERS

Stronger Cost  
Controls &  
Reduced Risks

### SUPPLIERS

Alignment of  
Prices with  
Patient  
Outcomes

### SOCIETY

Reduced  
Healthcare  
Spending &  
Better Overall  
Health

NEJM Catalyst ([catalyst.nejm.org](https://catalyst.nejm.org)) © Massachusetts Medical Society

- Trump elected in 2016
- Effects of MU so far:
  - 82% of EHR users report improved clinical decision-making
  - 92% report improvement in communication with other providers
  - 82% report a reduction in medication errors

*Article: Trump, EHR, & Meaningful Use: Continued Investment in Healthcare IT? Or the End of an Entitlement?*

<http://www.metacaresolutions.com/2017/01/trump-ehr-meaningful-use-continued-investment-healthcare-end-entitlement/>

- Republicans view government roles completely different than the Obama administration
  - Less regulatory oversight
  - Less government funded projects
  
- BUT: EHR experts argue it is unlikely a Trump presidency will slow down digital health momentum
  - Major effect will be on insurance ("Obamacare")
  
  - Movie and article at:  
<https://money.cnn.com/2018/01/20/news/economy/obamacare-trump-year-one/index.html>

- The American Hospital Association (AHA) is however asking Trump for deregulation of MU
  - Asks to cancel MU Stage 3 "so hospitals don't have to spend a fortune upgrading their electronic health records solely to meeting regulatory requirements"



- Deregulation
- More free market competition
- IH funding may cut and projects like the Cancer Moonshot, Precision Medicine (now named “AllOfUs”), and the Center for Medicare and Medicaid Innovation have been scaled back
- transition from fee for service to value based purchasing continue unmodified

- 8 ways Trump hurt Obamacare in the first year

<https://money.cnn.com/2018/01/20/news/economy/obamacare-trump-year-one/index.html>

- Is Meaningful User over (2018)?

<https://www.cerner.com/perspectives/is-meaningful-use-over>

CMS = Centers for Medicare and Medicaid Services

<https://www.cms.gov/>

- Trump Administration Announces MyHealthEData Initiative to Put Patients at the Center of the US Healthcare System

<https://www.cms.gov/newsroom/press-releases/trump-administration-announces-myhealthedata-initiative-put-patients-center-us-healthcare-system>

- Evolution of Healthcare Information Interoperability and What's Next for Payers (Oct. 2021)

<https://veradigm.com/veradigm-news/evolution-healthcare-interoperability-for-payers/>

- Patient Protection and Affordable Care Act

<https://www.britannica.com/topic/Patient-Protection-and-Affordable-Care-Act/Provisions>

- Essentially, restriction imposed under the Trump administration were reverted
- Article: Obamacare enrollment hits record high after Biden makes post-Trump tweaks to health insurance program

<https://www.cnbc.com/2022/01/18/obamacare-enrollment-hits-record-high-after-biden-makes-changes.html>

- US Government Lights FHIR Under Payers

<https://www.smilecdr.com/our-blog/us-government-lights-fhir-under-payers>



- <https://www.cms.gov/Regulations-and-Guidance/Guidance/Interoperability/index>

## Policy

- [CMS Interoperability and Patient Access Final Rule](#)
- [CMS Interoperability and Prior Authorization Proposed Rule](#)
- [Guidance for States](#)
- [Best Practices for Payers and App Developers](#)
- [Patient Privacy and Security Resources](#)

## Technical Standards

- [FHIR](#)
- [SMART/OAUTH 2](#)
- [Open ID Connect](#)
- [United States Core Data for Interoperability \(USCDI\)](#)

- <https://www.healthit.gov/isa/united-states-core-data-interoperability-uscdi>

## Version 1 versus Version 2 and Draft Version 3

- Exercise: which standard(s) is (or are) required for coding of
  - lab tests and results?
  - medications?
  - diagnoses?
  - units of measure?